Tax-Exempt Bond Bidding Marketplace

Working Brief — Everything Learned So Far

Status: **DRAFT** (brainstorming capture, not yet approved) | Date: July 20, 2026 | Working name: TBD (placeholder *CapStack*)

READ ME FIRST -- verification debt
Every statutory number, day-count, and compliance mechanic in this brief is marked **[VERIFY]** where certainty is not yet established. These are the load-bearing facts of the product -- none should be treated as true until confirmed against the IRC, Treasury regulations, or the relevant state QAP / agency rules. They are captured here as design intent and open questions, not as settled fact.

1. One-sentence thesis

A **compliant competitive-bidding marketplace for tax-exempt affordable-housing bond deals** — it runs the regulated bid events across a bond's life, matches **expiring tax-exempt capacity** to **shovel-ready projects**, and enforces the **compliance clocks** so nobody wins an illegal, expired, or non-competitive deal.

The moat is **encoded compliance + timing**, not matchmaking. A directory of public NOFAs/QAPs is nearly worthless (the PDFs are already public); the value is the software that (a) reads the 100-page rules, (b) runs a legally defensible competitive bid, and (c) back-schedules everything from a fixed bond date so the humans hit the deadline.

2. Domain primer (why this is hard)

The parties — the deal triangle

- **Issuer** — governmental / conduit issuer that *holds the tax-exempt capacity*. The capacity is on a clock.
- **Borrower** — the developer who borrows bond proceeds to **start the next project**.
- **Lender / bank** — actually funds the bonds (typically a direct purchase / private placement).

The perishable asset — tax-exempt bond capacity

- Private-Activity Bond (PAB) **volume cap** is allocated per state, per year (roughly per capita). **[VERIFY]** current per-capita figure.
- Financing **>=50%** of a project's aggregate basis + land with tax-exempt PABs auto-unlocks the **4% LIHTC** — no competitive QAP scoring fight (the "50% test"). **[VERIFY]** exact wording.
- **Carryforward:** unused cap can be carried forward and "captured." Your note: within **4 years** of original issue. My memory: statutory carryforward is **3 years** (IRC 146(f), Form 8328). **[VERIFY]**.
- **Recycling:** when a multifamily bond is repaid, the issuer can re-issue **recycled bonds without new volume cap** if done within **~6 months** (IRC 146(i)(6)). Catch: recycled bonds do **not** generate fresh 4% credits — cheap tax-exempt debt + cap preservation only. **[VERIFY]** window + the "issuer has interest / borrower does not" constraint.

The economics that create urgency

- If tax-exempt capacity **expires unused**, the next deal must use **taxable** debt at ~**+50 bps** (your figure; spread varies by market/credit).
- On a **$2.5M** issue that is ~**$12,500/yr** in extra interest — often the difference between a deal that pencils and one that dies.
- So the platform sells **saved basis points + capacity that would otherwise be wasted**, by assembling a closeable tax-exempt deal *before the clock runs*.

The 50-state problem

- LIHTC is allocated by each state's Housing Finance Agency under its own **Qualified Allocation Plan (QAP)** — different scoring, set-asides, deadlines.
- **California's QAP (CTCAC) is ~100 pages**, rewritten annually; every state differs; county/city NOFAs (e.g., **LACDA**) stack on top.
- 50 states x ~100 pages x annual revisions + local NOFAs = the rules no human holds in their head. Structuring these into machine-readable rules is the defensible work.

3. Users and business model (who pays)

Principle: charge the side that captures the most *durable* value and transacts *repeatedly* — the **banks** — not the one-off small developer. Never tax the holder of the scarce asset (the issuer); you want every dollar of expiring capacity *on* the platform.

Party	What they pay	Rationale
Issuer (supply)	Free	The reason the marketplace exists; add no friction to the scarce asset.
Bank (bidder)	Subscription (annual seat) + success fee on won deals	A single won $2.5M loan = 15-30 yrs of interest income; richest, most repeat participant.
Borrower (demand)	Free / minimal early	The liquidity you are courting; charge later via a small success fee, if at all.

Recommended fee stack

- **Free supply** (issuers list capacity/deals).
- **Bank subscription** (all-you-can-bid seat) — predictable revenue, the pros.
- **Success fee** on the winning bank when a deal closes (flat or small bps of the issue; e.g., 25 bps of $2.5M ~ $6,250) — aligns platform with *closed* deals.
- **Refundable bid deposit ($500-$1000)** as an anti-flake / seriousness gate — returned on a genuine bid, forfeited if a winner walks. Keeps the "pay to bid" instinct without taxing early liquidity.
- **Pay-per-bid** deferred until liquidity is healthy (optional later layer).

4. Core product -- the two engines

Everything rides on two shared engines, reused by every auction type.

4a. Countdown / back-schedule engine (the heartbeat)

You never pick the deadline — **the bond does.** Given a fixed anchor date, the platform generates the working schedule and drives the parties to it. Worked example (your scenario):

```
Apr 1   -- REDEMPTION (anchor) -- capacity freed / clock starts
Apr 8      platform posts the deal, notifies qualified bidders
Apr 22     bids due
May 6      award winner, generate compliance certificate
May 20     docs + counsel sign-off
Jun 1   -- TARGET: next deal closes / proceeds redeployed
```

[VERIFY] -- date direction (two clocks?)
You said GIC bidding must "complete 60 days *before* redemption," but the Apr 1 -> Jun 1 example lands 60 days *after* redemption. These are most likely **two different clocks** (a redemption-triggered recycle/deploy window vs. a bid-must-finish-before-X regulatory deadline). Must be pinned down before build.

4b. Compliance rulebook engine (the moat)

Ingests the 100-page rule sources into structured, queryable rules and validates every proposed pairing/award against them.

- **Sources:** state QAPs, agency NOFAs (LACDA / HCD / etc.), IRC + Treasury arbitrage and volume-cap rules.
- **Enforces:** eligibility (who may hold recycled authority), the 50% test, reserve/minimum, minimum number of bona-fide bidders, and every clock.
- **Never** lets the marketplace match an illegal or expired pairing.

5. Auction rules (the rulebook so far)

- **Sealed competitive bids.** Bona-fide competition is mandatory (partly a compliance requirement of the arbitrage safe harbor).
- **Reserve / minimum bid must be met** or **no award** (no forced sale below reserve). Direction depends on auction type.
- **No "buy-it-now."** A fixed-price shortcut would blow the bona-fide competition the safe harbor requires. Competition is the product.
- **Minimum number of bona-fide bidders** (**[VERIFY]** -- arbitrage GIC safe harbor typically requires >=3 bids from providers with no material interest; Treas. Reg. 1.148-5(d)(6)).
- **Refundable seriousness deposit** to bid; forfeited on winner walk-away.
- **Compliance certificate** auto-generated on award (documents the bona-fide bid for the file).

6. The two auction types

Type A — Loan-rate reverse auction (recommended v1 wedge)

- **What's bid:** the interest rate/terms to fund a posted deal. **Banks bid; lowest rate wins.**
- **Value:** directly compresses the +50 bps; fits the "start the next project" story.
- **"Minimum bid" here** = a reserve in the rate direction (a maximum acceptable rate, or minimum bid-validity standards); unmet -> no award.
- **Risk:** relationship-heavy; banks may resist disintermediation -> harder cold-start.

Type B — GIC / investment-agreement bidding (module 2)

- **What's bid:** the yield on a Guaranteed Investment Contract for un-spent bond proceeds/reserves. **Providers bid; highest qualifying yield wins.**
- **"Minimum bid" here** = a reserve/floor yield the winner must at least meet (safe-harbor benchmark). Natural fit for the "minimum must be met" rule.
- **Why it's a tempting alternative wedge:** most standardized/rule-bound -> easiest to make provably correct; recurs on every deal; there is already a manual market (bidding agents) that pays for it -> clearer willingness-to-pay and a known incumbent to beat.
- **Downside:** it's the "plumbing," not the headline origination story.

Both auctions share the two engines. Choosing the wedge decides which flow ships first, not the architecture.

7. v1 scope (proposed) vs. roadmap

v1 -- the wedge (assume Type A unless redirected)

- One auction type (loan-rate reverse auction).
- Countdown engine + compliance rulebook for **one jurisdiction first** (proposed: California / CTCAC, largest LIHTC market and the hardest 100-page QAP). **[VERIFY]** CA vs. your home market (NM?).
- Bank subscription + refundable deposit + success fee.
- LLM-assisted, human-reviewed rule ingestion for that one jurisdiction -- not yet automated 50-state.

Roadmap -- later modules, not v1

- Type B (GIC bidding).
- Additional states (clone the ingestion pipeline; breadth after depth).
- Developer-side tooling / QAP fit-scoring.
- Capacity matching across a $100M issuer pool -> many $2.5M deals.
- Full three-party deal-assembly workflow to close.

8. Data model sketch (first cut)

- **Issuer** -- id, jurisdiction, capacity holdings.
- **Capacity** -- type (fresh cap / carryforward / recycled), amount, jurisdiction, expiry clock, eligibility constraints.

- **Bond** -- anchor dates (issue, redemption, maturity), status.
- **Project (Deal)** -- borrower, location, unit count, AMI targets, size, readiness.
- **Auction** -- type (A/B), reserve, min-bidders, open/close dates (derived by countdown engine), status.
- **Bid** -- bidder, sealed value (rate or yield), deposit status, validity flags.
- **Party** -- issuer / borrower / bank / GIC-provider, KYC/eligibility.
- **ComplianceRule** -- source doc, machine-readable predicate, jurisdiction, effective dates.
- **Clock** -- anchor date, rule, computed deadlines, notifications.
- **Award + ComplianceCertificate** -- winner, generated evidence-of-competition.

9. Assumptions and OPEN QUESTIONS

These are the decisions that change the build. Your answers here unlock the implementation plan.

1. **Wedge:** defaulting to Type A (loan-rate). Confirm -- or switch to Type B (GIC) for a faster provable v1.
2. **Bidders in Type A** = banks bidding rate. Confirm it isn't developers bidding for capacity (different auction, different payer).
3. **First jurisdiction** = California / CTCAC. Confirm vs. NM or wherever your relationships are.
4. **Date direction:** two clocks? Resolve "60 days before" redemption vs. the Apr 1 -> Jun 1 "60 days after."
5. **Carryforward window:** 3 years (my memory) vs. 4 years (your note). [VERIFY].
6. **Recycling:** 6-month window + the "issuer has interest / borrower does not" rule. [VERIFY] exact constraint.
7. **Min-bidders / safe harbor:** >=3 bids? Which safe harbor governs each auction type? [VERIFY].
8. **Customer #1** you can actually call next week -- a specific issuer, bank, or developer? (Determines go-to-market.)

10. Explicitly OUT of scope (YAGNI, for now)

- Automated 50-state rule ingestion on day one (depth-first on one state instead).
- Two-sided cold-start of the full capital marketplace.
- Developer QAP fit-scoring product.
- Anything that reproduces public PDFs as a "directory" (no moat).

11. Key risks

- **Regulatory correctness** -- the value prop is compliance; a wrong clock or eligibility rule is a legal liability, not just a bug. -> Human-in-the-loop review of every ingested rule; counsel sign-off gate.
- **Cold-start liquidity** -- no bids = no marketplace. -> Start where one issuer's pool creates its own liquidity; deposit-not-fee to bid.

- **Disintermediation resistance** (Type A) -- banks/agents may resist. -> Type B (GIC) has an existing paid manual market that's easier to convert.
- **Verification debt** -- every [VERIFY] above is a landmine if shipped unverified.

12. Next steps

1. Review this brief and correct Section 9 (open questions) + any [VERIFY] facts you already know.
2. Lock the wedge (A vs B) and the first jurisdiction.
3. Then: turn it into a detailed, step-by-step implementation plan for v1.